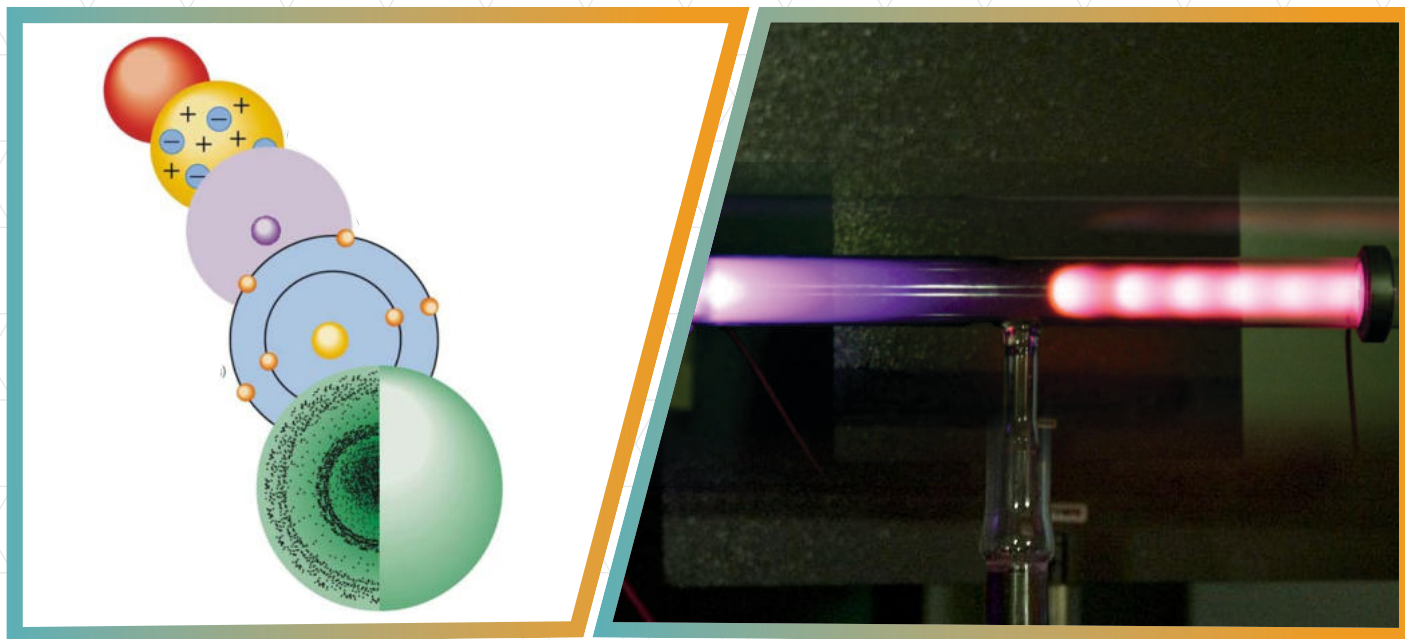




UNIT 02

ATOMIC STRUCTURE



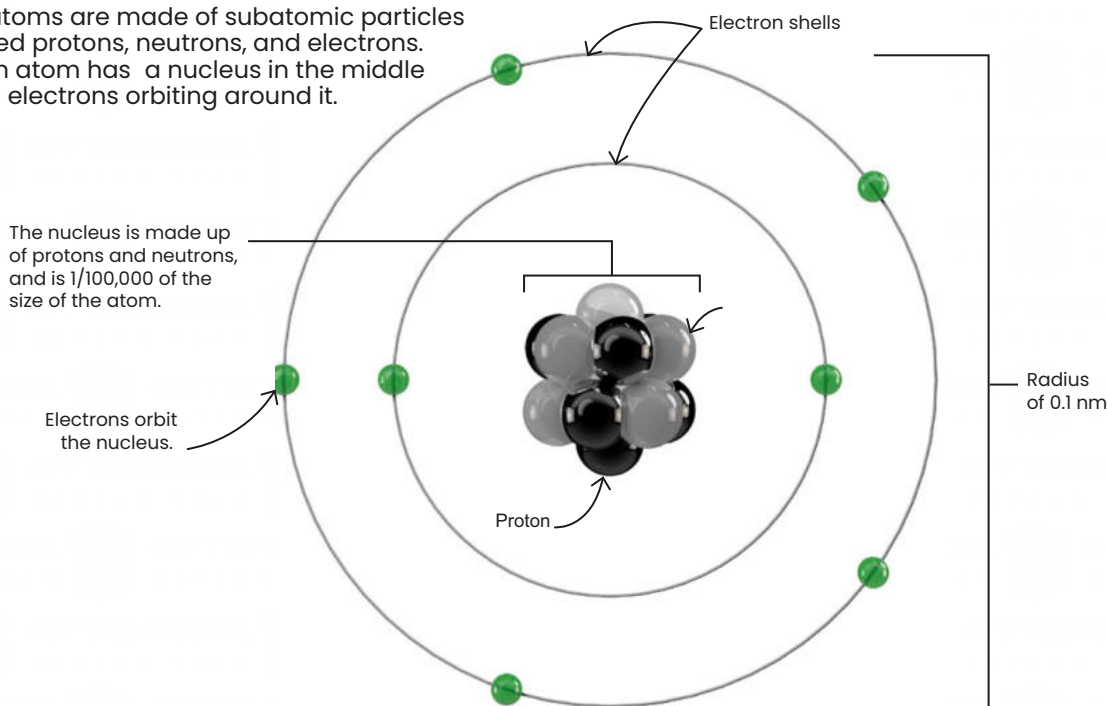
INTRODUCTION

John Dalton (1805) considered that **all matter was composed of small particles called atoms^[1]**. He visualized the atom as a hard-solid individual particle incapable of subdivision.

At the end of the nineteenth century there accumulated enough experimental evidence to show that the atom is made of still smaller particles. These subatomic particles are called the fundamental particles.

Atomic structure

All atoms are made of subatomic particles called protons, neutrons, and electrons. Each atom has a nucleus in the middle with electrons orbiting around it.



SHORT QUESTIONS

1) Define sub-atomic particles?

The number of subatomic particles now known is very large. For us, the three most important are the proton, neutron and electron. And they can be defined as:

A) DEFINITION OF AN ELECTRON

Those sub-atomic particles which revolve around the nucleus in their energy states, shells or orbit.

B) DEFINITION OF PROTON

Those sub-atomic particles which exist in the nucleus of an atom and bears a positive charge.

C) DEFINITION OF NEUTRON

Those sub-atomic particles which exist in the nucleus of an atom and it has no charge because it is electrically neutral.

2) What is isotope?

DEFINITION OF ISOTOPE (GREEK, ISOS = SAME; TOPOS = PLACE)

The atoms of an element which have the same number of protons and different number of neutrons are called Isotopes.

1 The word atom is derived from a Greek word **ATOMOS** means indivisible which was first describe by the Greek philosopher Democritus. (Textbook of chemistry for grade 9, Sindh text board, page # 34)

Previous Year's Questions



How does electron differ from neutron?

[2025]

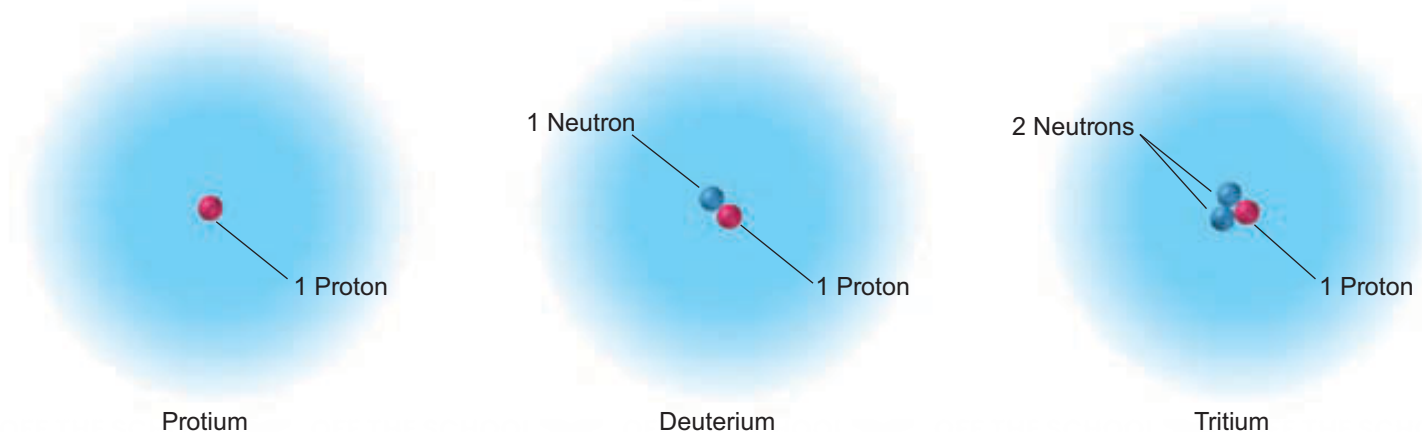
Alternatively, isotopes may be defined as: **The atoms of an element which have the same atomic number but different atomic masses or mass numbers.**

3) Draw the structure of isotopes of hydrogen to justify the definition of isotope?

ISOTOPES OF HYDROGEN

The naturally occurring isotopes of hydrogen are three, present in different abundances. The three isotopes of hydrogen are named as **(i) Protium (ii) Deuterium (iii) Tritium**. Each one of them has 1 proton and 1 electron, but number of neutrons are different.

The isotopes of hydrogen are represented as:



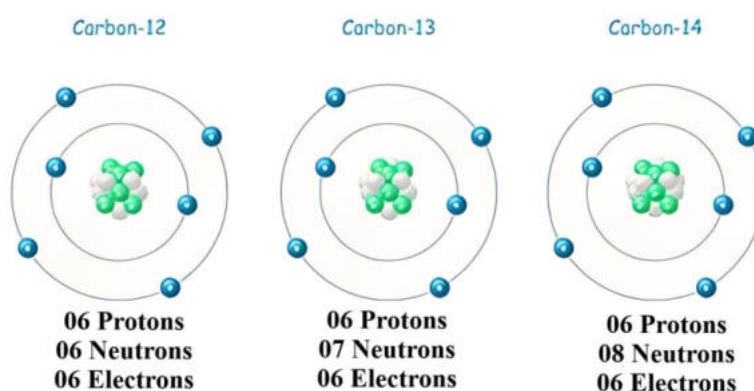
4) Write the isotopes of carbon?

ISOTOPES OF CARBON

There are total three isotopes of carbon exist, two are stable isotopes of carbon ^{12}C and ^{13}C and one is radioactive isotope ^{14}C .

The isotope ^{12}C is present in abundance of 98.9 %, while ^{13}C and ^{14}C are both present only 1.1 % in nature. All of them have the same number of protons and electrons but differ in number of neutrons. They are represented as follows:

Isotopes of carbon



5) What do you know about the isotopes of chlorine and uranium?

ISOTOPES OF CHLORINE

There are two isotopes of chlorine found in nature i.e. $^{35}_{17}\text{Cl}$, $^{37}_{17}\text{Cl}$ 75% and 25% respectively.

ISOTOPES OF URANIUM

There are three isotopes of uranium all are radioactive i.e. $^{234}_{92}\text{U}$, $^{235}_{92}\text{U}$, $^{238}_{92}\text{U}$ is found in nature nearly 99%.

NAME AND SYMBOLS OF ELEMENTS

The name of the elements is given by its inventors;

- 1) Some are named after the place like an element Berkelium (Bk) after its discovery in Berkeley (University of California, Berkeley).
- 2) Some are based on planet, like element Mercury (Hg) named after the planet mercury.
- 3) Some given to honor the great scientist, like curium (Cm) after Madame Curie.

Atomic Number, Mass Number, Number of Protons and Neutrons of the isotopes of Hydrogen, Carbon, Chlorine and Uranium:

SYMBOL	ATOMIC NUMBER	MASS NUMBER	NO. OF PROTON	NO. OF NEUTRON
^1H	1	1	1	0
^2H	1	2	1	1
^3H	1	3	1	2
^{12}C	6	12	6	6
^{13}C	6	13	6	7
^{14}C	6	14	6	8
^{35}Cl	17	35	17	18
^{37}Cl	17	37	17	20
^{234}U	92	234	92	142
^{235}U	92	235	92	143
^{238}U	92	238	92	146

6) Atomic number and mass number? Both terms are same or not?

ATOMIC NUMBER (Z):

The number of protons present in the nucleus is called Atomic number.

MASS NUMBER (A):

The mass number of an atom is the total number of protons and neutrons in its nucleus.

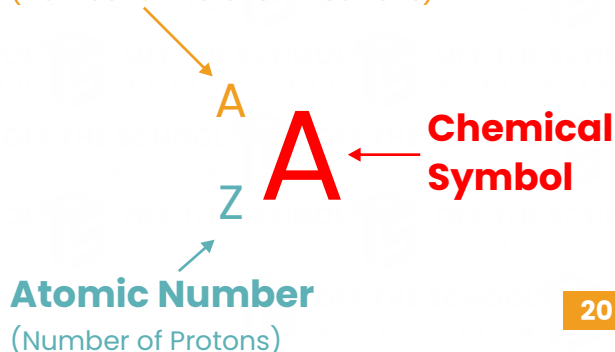
DIAGRAMMATICAL REPRESENTATION WITH EXPLANATION:

Now we illustrate the key points from the diagram:

- Mass number is denoted by "A" and atomic number is represented by "Z".
- 'A' shown in the diagram is any element like C, N, O and so on...
- Atomic number represents the protons.
- The number of proton and electron is equal to Z in an isolated or neutral atom.
- If we see any element like $[_Z\text{X}^A = {}_6\text{C}^{12}]$ so the subscript (base) represent Z and superscript (power) represent A.

Mass Number

(Number of Protons + Neutrons)



7) What are energy levels?

DEFINITION

"They are the possible and suitable path for electron around the nucleus in which they orbiting. Energy levels are also called shells or stationary states."

EXPLANATION

They are represented by 'n'; these shells are named as K, L, M, N, O, P with quantum numbers $n = 1, 2, 3, 4, 5, 6$ respectively. These shells have definite amount of energy. And energy increases continuously as the number of shell increases from lower to higher (K to P).

8) Highlight the term sub-energy levels or sub-shells?

The splitting of energy-levels are sub-energy levels. Actually, the energy level is divided into sub-energy levels which named as s, p, d & f^[1]. It is also known as orbitals.

LONG QUESTIONS

1. Describe discharge tube experiment with labelled diagram and state what were the justification of and conclusion of J.J Thomson in this respect? (2022)

CATHODE RAYS – THE DISCOVERY OF ELECTRON

I) INTRODUCTION:

The knowledge about the electron was derived as a result of the study of the electric discharge in the discharge tube (J.J. Thomson, 1896).

II) APPARATUS SETUP:

The discharge tube consists of a glass tube with metal electrodes fused in the walls and gas filled in it.

Through a glass side-arm air can be drawn with a pump (Vacuum pump).

✓ The electrodes are connected to a source of high voltage (10,000 Volts).

✓

III) WORKING:

When high voltage current was passed through the gas, shiny rays were emitted from the cathode which travel towards the anode as shown in diagram above. These rays were given the name of "cathode rays" as these were originated from the cathode.

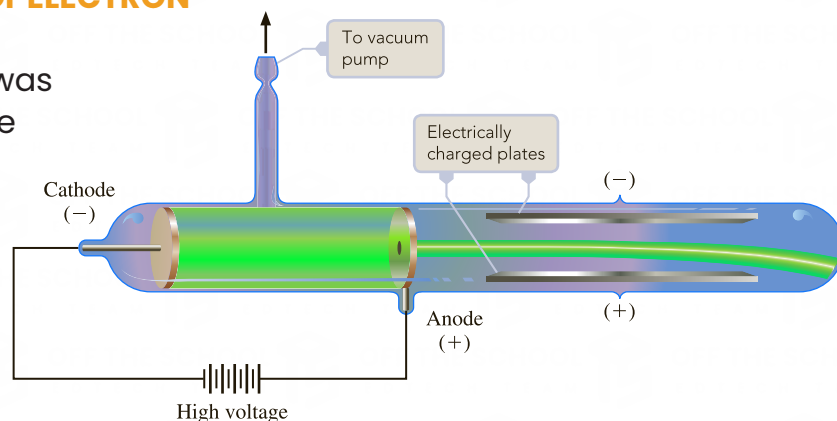
PROPERTIES OF CATHODE RAYS:

The cathode rays were studied in detail and their properties were determined, which are given below:

(i) These rays travel in straight lines perpendicular to the cathode surface.

(ii) They can cast a sharp shadow of an opaque object if placed in their path.

(iii) They are deflected towards positive plate in an electric field showing that they are negatively charged.



Previous Year's Questions



Write down three properties of cathode rays.

[2023]

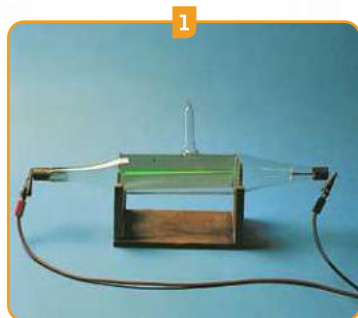
Green beam shows the deflection of cathode ray in electric field

(iv) They raise temperature of the body on which they fall.

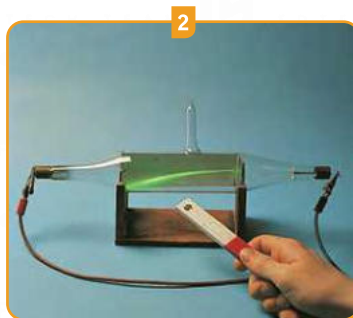
(v) JJ. Thomson discovered their charge/mass (e / m) ratio.

(vi) Light is produced when these rays hit the walls of the discharge tube.

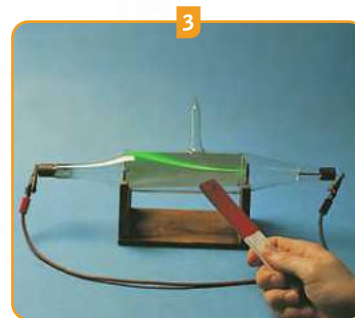
(vii) It was found that the same type of rays were emitted no matter which gas and which cathode was used in the discharge tube.



The cathode-ray beam travels from right to left. It is visible where it falls on a zinc-sulfide screen.



The beam of negative particles bends downward as the south pole of the magnet is brought toward it.



When the magnet is turned around, the beam bends in the opposite direction.

CONCLUSION

Different scientists tried different discharge tubes with different electrodes and different gases but they found the same results.

Experiment shows that these particles could be produced by any kind of material. Cathode rays consisting of particle which are known as ELECTRONS.^[1]

2. Briefly mention the discovery of proton with diagram? Or what is perforated cathode experiment give briefing? Or write the discovery and properties of that rays which is moving from positive terminal to negative terminal when electric current is passed through the discharge tube?

DISCOVERY OF PROTON

INTRODUCTION:

In 1886 Eugen Goldstein used a discharge tube with a hole in the cathode said to be perforated cathode.

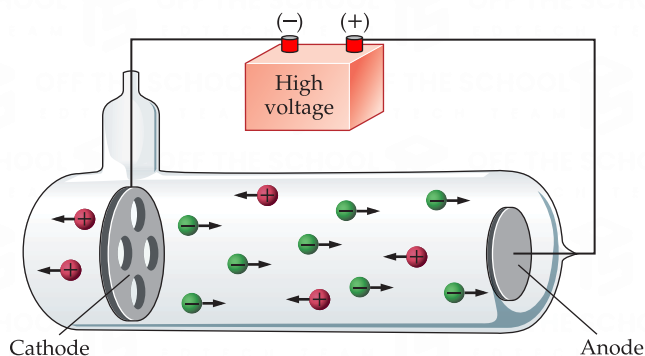
APPARATUS:

Same apparatus were used as in cathode rays' experiment.

- ✓ Discharge tube filled with gas.
- ✓ High voltage battery.
- ✓ Vacuum pump.
- ✓ Electrodes in which cathode is perforated.

WORKING:

He observed that while cathode rays were streaming away from the cathode, there were coloured rays produced simultaneously which passed through the perforated cathode and caused a glow on the wall opposite to the anode. Thomson studied these rays and showed that they consisted of particles carrying a positive charge. He called them Positive rays.



¹ Dutch Physicist H.A. Lorentz named them **Electrons**.

PROPERTIES OF POSITIVE RAYS OR CANAL RAYS:

The properties of these rays were as following:

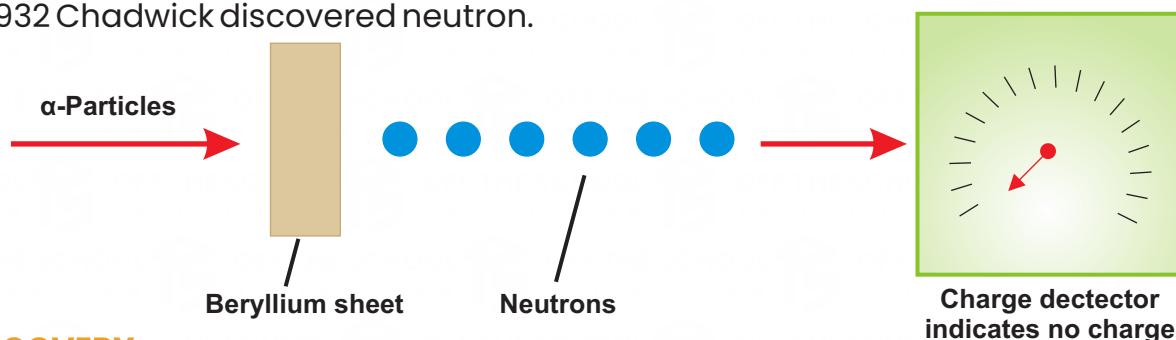
- (i) These rays travel in straight lines in a direction opposite to the cathode rays.
- (ii) Their deflection in electric and magnetic field proved that these are positively charged.
- (iii) The nature of canal rays depends upon the nature of gas, present in the discharge tube.
- (iv) They possess mass many times the mass of an electron.
- (v) They cause fluorescence in zinc sulfide.

3. Explain the discovery of neutron and their properties?

DISCOVERY OF NEUTRONS

INTRODUCTION:

In 1932 Chadwick discovered neutron.



DISCOVERY:

When he bombarded alpha particles on a beryllium sheet. So alpha-Particles hit directly at beryllium sheet and eject neutrons whereby the electric charge detector remains unaffected.

He observed that highly penetrating radiations were produced. These radiations were called neutrons.



PROPERTIES OF NEUTRON:

Properties of neutron are as following:

- (i) Neutrons carry no charge i.e. they are neutral.
- (ii) They are highly penetrating.
- (iii) Mass of these particles was nearly equal to the mass of a proton.

PARTICLE	SYMBOLS	RELATIVE ELECTRIC CHARGES	MASS NUMBER	RELATIVE MASS(U*)	ACTUAL MASS (KG)
Electron	e^- , ${}_{-1}^0e$	-1	0	0.000 5486	9.109×10^{-31}
Proton	p^+ , ${}_1^1\text{H}$	+1	1	1.007 276	1.673×10^{-27}
Neutron	n^0 , ${}_0^1\text{n}$	0	1	1.008 665	1.675×10^{-27}

4. How did Rutherford present the atomic model? Describe his experiment in detail and explain the atomic model presented by him. [2025]

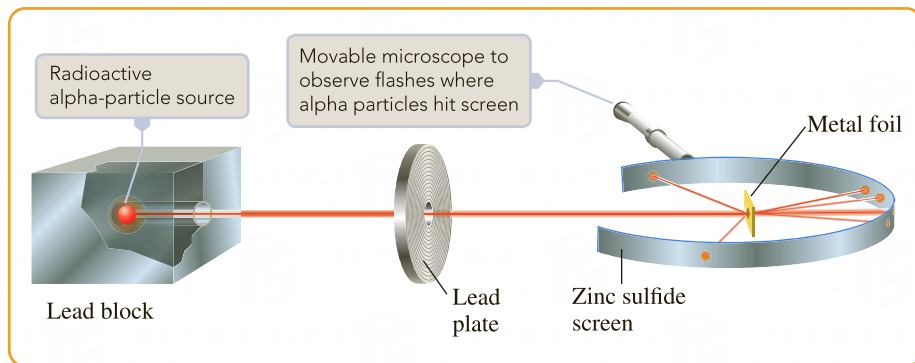
RUTHERFORD'S ATOMIC MODEL – THE NUCLEAR ATOM

INTRODUCTION:

In 1909 Rutherford performed their historic Alpha Particle Scattering Experiment.

Alpha-particle scattering from metal foils

Alpha radiation is produced by a radioactive source and formed into a beam by a lead plate with a hole in it. (Lead absorbs the radiation.) Scattered alpha particles are made visible by a zinc sulfide screen, which emits tiny flashes where particles strike it. visible by a zinc sulfide screen, which emits tiny flashes where particles strike it.



APPARATUS:

- ✓ Radium^[1] a source of alpha particles.
- ✓ Thin gold foil.
- ✓ A lead box for emitted radiations.
- ✓ Fluorescent or luminescent screen made up of ZnS.

LINK TO LEARNING

View this simulation (<http://openstaxcollege.org/l/16Rutherford>) of the Rutherford gold foil experiment. Adjust the slit width to produce a narrower or broader beam of a particles to see how that affects the scattering pattern.

LINK 01

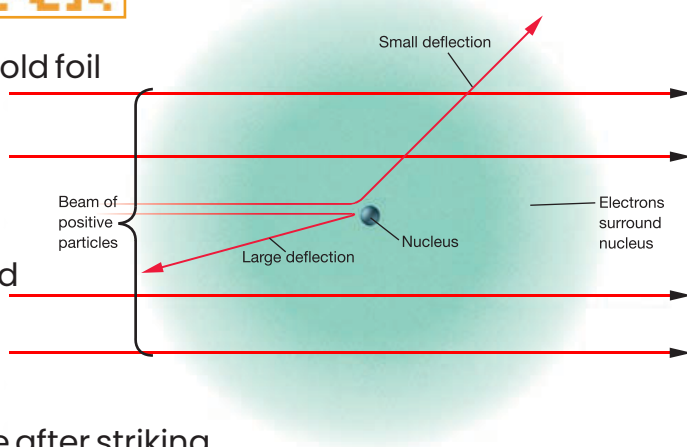


EXPERIMENT:

Rutherford fired a particles at a piece of thin gold foil and detected where those particles went.

OBSERVATIONS:

- Most of the α -particles (positive particles) passed straight through the foil and produced illumination^[2] on screen.
- Some α -particles show deviation (slightly deflected from their path).
- Very few α -particles ($1/8000$) are rebound after striking.



CONCLUSION:

- (i) The volume occupied by an atom must consist of a large amount of empty space.
- (ii) A small, relatively heavy, positively charged body, the nucleus, must be at the center of each atom.

¹ A beam of high-speed, positively charged alpha particles (α particles) that were produced by the radioactive decay of radium (a radioactive element).

² If you want to know the illumination then look at your text book page 40 (Box of "Do you know?")

LINK TO LEARNING

The Rutherford scattering simulation (<http://openstaxcollege.org/l/16PhetScatter>) allows you to investigate the differences between a “plum pudding” atom and a Rutherford atom by firing a Particles at each type of atom. (Chemistry L.R OpenStax College, pg # 77, 11 March 2015)

LINK 02



MAIN POSTULATES:

- (1) Atom has a nucleus which contains practically the overall mass of the atom, leaving the rest of the atom almost empty.
- (2) The entire positive charge of the atom is located on the nucleus, while electrons were distributed in vacant (empty) space around it.
- (3) The electrons were moving in orbits or closed circular paths around the nucleus like planets around the sun.

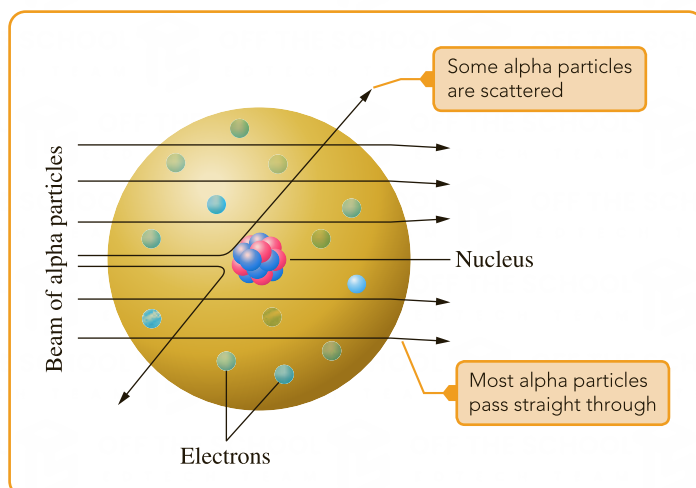
Rutherford described finding these results: “It was quite the most incredible event that has ever happened to me in my life. It was almost as incredible as if you fired a 15-inch shell at a piece of tissue paper and it came back and hit you.” (Ernest Rutherford, “The Development of the Theory of Atomic Structure,” ed. J. A. Ratcliffe, in Background to Modern Science, eds. Joseph Needham and Walter Pagel, (Cambridge, UK: Cambridge University Press, 1938), 61–74)

WEAKNESS OR DRAWBACKS OR DEFECTS OF THE RUTHERFORD ATOMIC MODEL:

The assumption that electrons were orbiting around the nucleus was unfortunate. If an electron radiates energy, its speed will decrease and it will go into spiral motion, finally falling into the nucleus. This does not happen actually as then the atom would be unstable which it is not. This was the chief weakness of Rutherford’s Atomic Model.

► Representation of the scattering of alpha particles by a gold foil

Most of the alpha particles pass through the foil barely deflected. A few, however, collide with gold nuclei and are deflected at large angles. (The relative sizes of nuclei are smaller than can be drawn here.)



5. How did bohr's represented his atomic model by the modification of Rutherford's model? Also mention the limitations of bohr's model?

BOHR'S ATOMIC MODEL

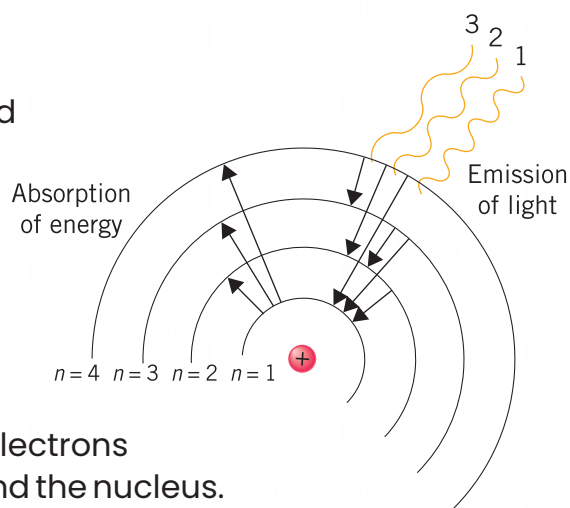
INTRODUCTION:

Neil Bohr was a Danish physicist who joined Rutherford in 1912 for his post-doctoral research. He presented his model of atom in 1913. He also explained the line spectrum of hydrogen atom.

MAIN POSTULATES OF BOHR'S:

The Bohr's atomic model was based upon the following postulates:

- ▶ The hydrogen atom consists of a tiny nucleus and electrons are revolving in one of circular orbits of radius 'r' around the nucleus.
- ▶ Each orbit has a fixed energy that is quantized.

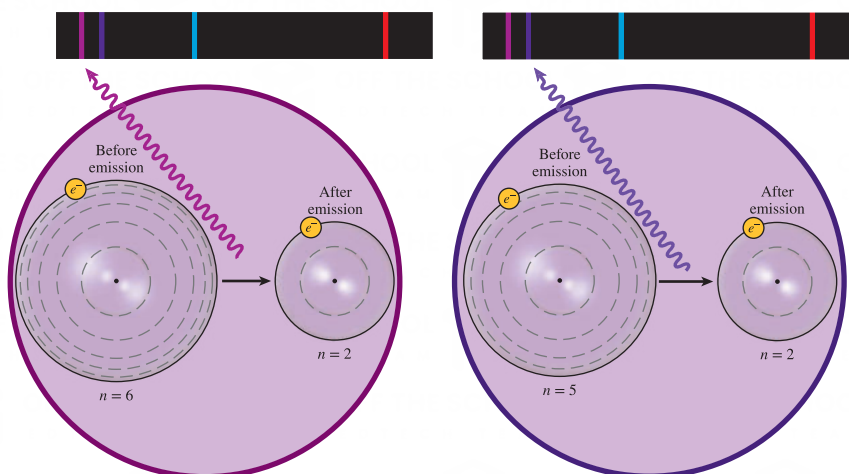


CIRCULAR ELECTRON ORBITS OR STATIONARY ENERGY LEVELS IN AN ATOM:

- ▶ As long as electron remains in a particular orbit, it does not radiate or absorb energy in the form of photon. The energy is emitted or absorbed only when an electron jumps from one orbit to another.
- ▶ When an electron jumps from lower orbit to higher orbit, it absorbs energy and when it jumps from higher orbit to lower orbit it radiates energy. This change in energy, ΔE is given by following Planck's equation:

$$\Delta E = E_2 - E_1 = h\nu$$

- ▶ Where, 'h' is Planck's constant equal to 6.63×10^{-34} J.s, and 'v' is frequency of light. Electron can revolve only in orbits of a fixed angular momentum (mvr), given as:
Angular momentum = $mvr = n \frac{h}{2\pi}$ Where 'n' is the quantum number or orbit number having values 1, 2, 3 and so on.



LIMITATIONS OF BOHR'S ATOMIC MODEL:

Bohr's model failed to explain

- ▶ Zeeman effect.
 - ▶ Stark effect.
 - ▶ Heisenberg's uncertainty principle.
 - ▶ Spectra of large atoms.
- Monoelectronic species like H^{+1} , Li^{+2} etc.

6. Discuss the concept of electronic configuration?

DEFINITION

"The distribution of electrons among the atomic orbitals in the atom."

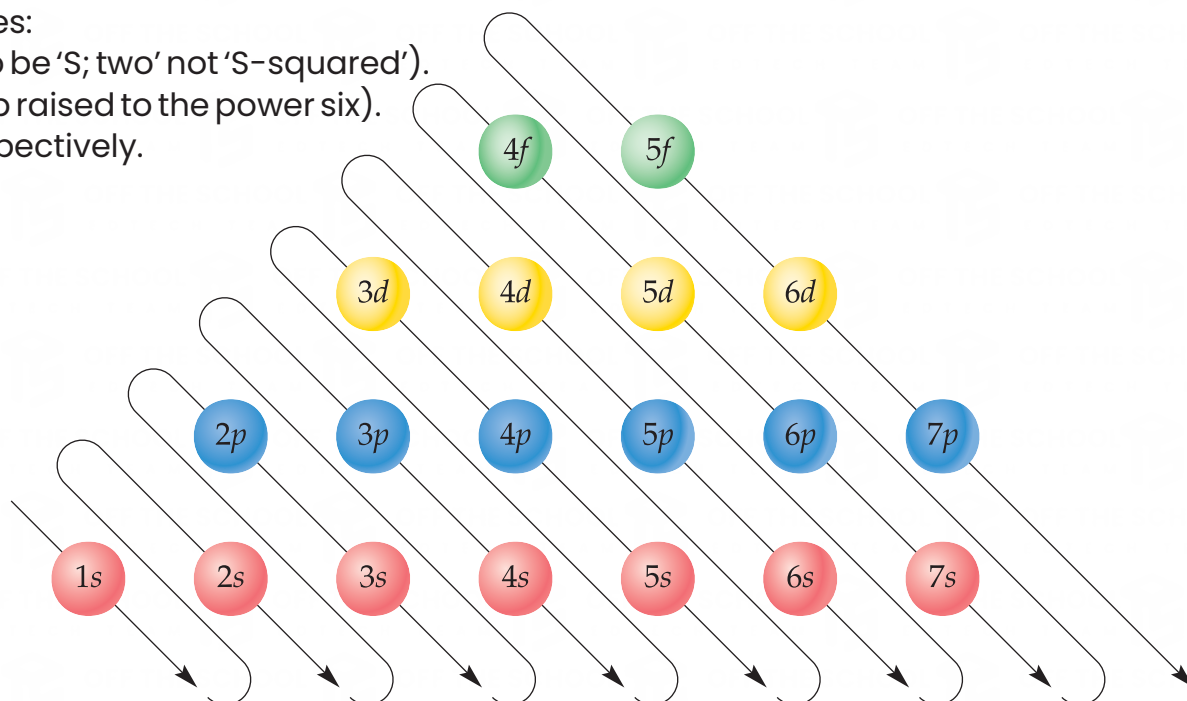
- ▶ 2 electrons in 's' subshell.
- ▶ 6 electrons in 'p' subshell.
- ▶ 10 electrons in 'd' subshell.
- ▶ 14 electrons in 'f' subshell.

So, it becomes:

s^2 (it is said to be 'S; two' not 'S-squared').

p^6 (p; six not p raised to the power six).

d^{10} and f^{14} respectively.



▲ An order-of-filling chart for determining the electronic configuration of atoms.

7. Explain the shell concept and also express the maximum capacity of each shell for electrons?

This concept was given by Neil Bohr that the distribution of electrons among the energy levels or shells.

Shell or principal quantum number is represented by 'n'.

So first Shell is K Shell which is nearest to Nucleus having n value equal to 1.

MAXIMUM ACCOMMODATION OF ELECTRON IN A SHELL

It can be calculated by the formula : $2n^2$

▶ For 1st K-shell ($n = 1$)

By formula $2n^2 \longrightarrow 2(1)^2 = 2e$.

▶ For 2nd Shell or L Shell ($n = 2$)

By formula $2n^2 \longrightarrow 2(2)^2 = 2 \times 4 = 8e$

▶ For 3rd shell or M Shell ($n = 3$)

By formula $2n^2 \longrightarrow 2(3)^2 = 2 \times 9 = 18e$

▶ For 4th Shell or N Shell ($n = 4$)

By formula $2n^2 \longrightarrow 2(4)^2 = 2 \times 16 = 32e$

Let's have a look on the electronic configuration of first 18 elements:

ELEMENTS	SYMBOLS	ATOMIC NUMBER	ELECTRONIC CONFIGURATION
Hydrogen	H	1	1s ¹
Helium	He	2	1s ²
Lithium	Li	3	1s ² , 2s ¹
Beryllium	Be	4	1s ² , 2s ²
Boron	B	5	1s ² , 2s ² , 2p ¹
Carbon	C	6	1s ² , 2s ² , 2p ²
Nitrogen	N	7	1s ² , 2s ² , 2p ³
Oxygen	O	8	1s ² , 2s ² , 2p ⁴
Fluorine	F	9	1s ² , 2s ² , 2p ⁵
Neon	Ne	10	1s ² , 2s ² , 2p ⁶
Sodium	Na	11	1s ² , 2s ² , 2p ⁶ , 3s ¹
Magnesium	Mg	12	1s ² , 2s ² , 2p ⁶ , 3s ²
Aluminium	Al	13	1s ² , 2s ² , 2p ⁶ , 3s ² , 3p ¹
Silicon	Si	14	1s ² , 2s ² , 2p ⁶ , 3s ² , 3p ²
Phosphorus	P	15	1s ² , 2s ² , 2p ⁶ , 3s ² , 3p ³
Sulphur	S	16	1s ² , 2s ² , 2p ⁶ , 3s ² , 3p ⁴
Chlorine	Cl	17	1s ² , 2s ² , 2p ⁶ , 3s ² , 3p ⁵
Argon	Ar	18	1s ² , 2s ² , 2p ⁶ , 3s ² , 3p ⁶

8. Describe wave particle duality of electron of de-Broglie's hypothesis?

In 1923, Louis de Broglie^[1], a French scientist, hypothesized that: "every moving matter has a wave associated with it." He also calculated that these properties are especially significant in subatomic particles.

De Broglie combine the Einstein and Planck equations and argued that if

$$\therefore E = h\nu$$

Where,

- E = energy
- h = plank'constant
- ν = frequency of light

And,

$$\therefore E = mc^2$$

Where;

- E = energy
- m = mass,
- c = speed of light (3×10^8 m/s)

then,

$$E = E$$

$$h\nu = mc^2$$

$$\left[\therefore \frac{c}{\lambda} = \nu \right]$$

$$h \frac{c}{\lambda} = mc^2$$

$$\lambda = \frac{h}{mc}$$

$$\left[\therefore mv = mc = p = \text{momentum} \right]$$

$$\lambda = \frac{h}{p}$$

¹ Only in France it says as (dee-brag-lee) but the rest of France pronounce this as "de Brail" (de-Burh-y). Actually It's an Italian of origin : de Broglie pronounced Bro-lyah..

9. Describe the Schrodinger atomic model?

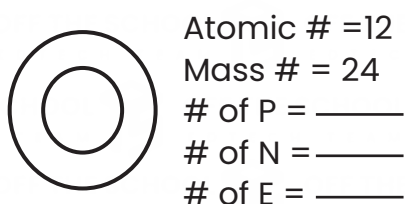
Beginning in January 1926, Erwin Schrödinger (shrow·duhng·uh), an Austrian scientist, applied the principles of wave mechanics to atoms and developed the quantum mechanical model of the atom.

The quantum mechanical model is both mathematical and conceptual. It keeps the quantized energy levels that Bohr introduced. In fact, it uses four quantum numbers to describe electron energy. These refer to

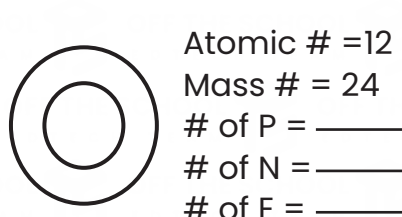
- (1) the principal energy levels.
- (2) the sublevel.
- (3) the orbital.
- (4) the number of electrons in an orbital.

ATOMIC STRUCTURE WORKSHEET

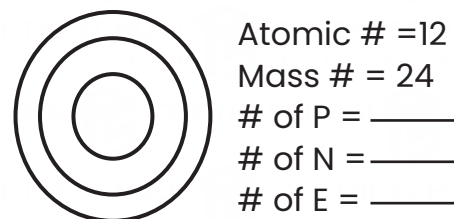
1. Name the three particles of the atom and their respective charges are:
 - a. _____
 - b. _____
 - c. _____
2. The atomic number tells you the number of _____ in one atom of an element. It also tells you the number of _____ in a neutral atom of that element. The atomic number gives the “identity” of an element as well as its location on the Periodic Table. No two different elements will have the _____ atomic number.
3. The _____ of an element is the total number of protons and neutrons in the _____ of the atom.
4. Name the element which has the following numbers of particles. Be specific. (Include charges and mass numbers where possible.)
 - (a) 26 electrons, 29 neutrons, 26 protons _____
 - (b) 53 protons, 74 neutrons _____
 - (c) 2 electrons (neutral atom) _____
 - (d) 20 protons _____
 - (e) 86 electrons, 125 neutrons, 82 protons (charged atom) _____
 - (f) 0 neutrons _____
5. If you know **ONLY** the following information can you always determine what the element is? (Yes/No).
 - (a) number of protons ().
 - (b) number of neutrons ().
 - (c) number of electrons in a neutral atom ().
 - (d) number of electrons ().
6. Calculate the missing information



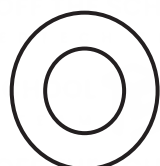
Li



Ne

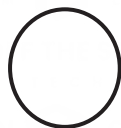


Mg



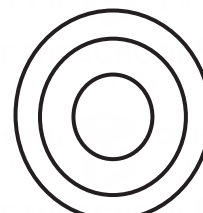
Cl

Atomic # = 12
Mass # = 24
of P = ____
of N = ____
of E = ____



He

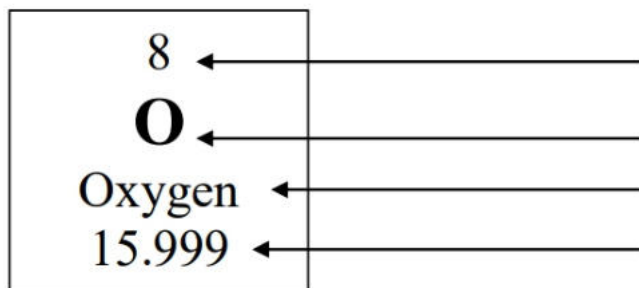
Atomic # = 12
Mass # = 24
of P = ____
of N = ____
of E = ____



Si

Atomic # = 12
Mass # = 24
of P = ____
of N = ____
of E = ____

7. Label the information provided in the periodic table.



8. You can become more familiar with the atomic structure of some elements by completing the chart. You have been given enough information to fill in all the blanks. (There is no need to use a periodic table.)

Substance	Symbol	Z	A	Number of Protons	Number of Neutrons	Number of electrons
(a) Helium		2	4			
(b) Magnesium		12			12	
(c) Zinc		30	65			
(d) Bromine			80			35
(e) Aluminum				13	14	
(f) Uranium					146	92
(g) Sodium		11			12	
(h) Hydrogen		1	1			
(i) Calcium			40	20		
(j) Silver				47	61	
(k) Mercury		80	201			
(l) Iodine			127	53		
(m) Potassium					20	19

9. How many electrons can each level hold? 1st = ____ 2nd = ____ 3rd = ____
10. What term is used for the electrons in the outermost shell or energy level?
